

SODA: Supervised Option Discovery for Dynamic Action Chunking

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Motivation

Humans plan by intent: actions end when a sub-goal completes, not on a fixed timer. Action chunking policies (Diffusion Policy, ACT) have no such concept — they predict a fixed-length chunk of T_p steps and execute $T_a \leq T_p$ before replanning. Two fundamental limitations arise:

- **Fixed prediction horizon T_p :** robot tasks decompose into semantic subtasks (*reposition*, *push*, *align*) with *variable* durations. A fixed T_p under-plans long subtasks and over-plans short ones — misscoping the chunk degrades even the first T_a steps executed.
- **Consistency–reactivity tradeoff [BID, 2024]:** large T_a gives smooth execution but cannot react to perturbations; small T_a enables reactivity but causes jitter from constant replanning. No fixed value is optimal across all task phases.

SODA resolves both: π_{high} selects *semantic options* that scope T_p , and learned β triggers replanning when invalid (in-option or new option).

Related Work

Action Chunking & BID

ACT / Diffusion Policy [Chi '23, Zhao '23] use fixed T_p and fixed T_a , mismatching variable subtasks and the consistency–reactivity tradeoff. BID [Xie '24] improves test-time replanning but not semantic scoping of T_p .

Horizons, Skills, Options

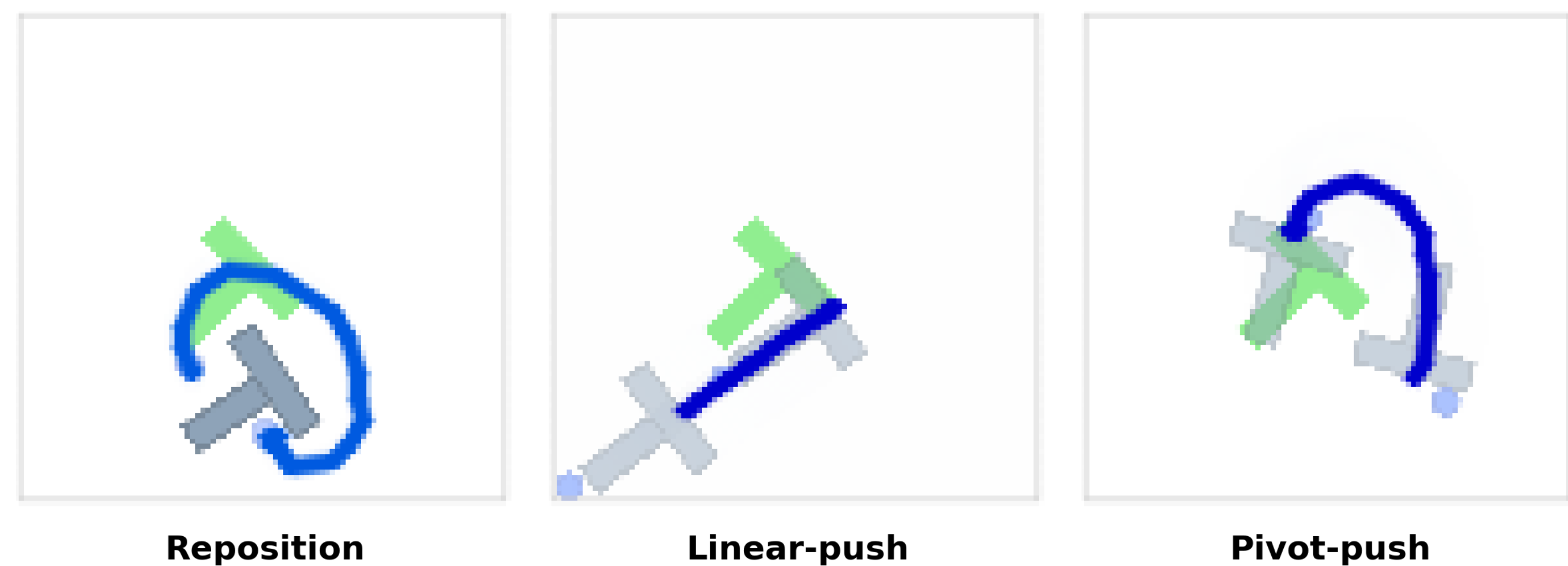
Pfrommer et al. add a horizon channel (min-time RL, not demo semantics). ComPILE / LOVE [Kipf '19, Jiang '22] segment demos without labels tuned for control; weak supervision (TACO, sketches) lacks VLM grounding [Sutton '99].

Gap

SODA unifies (1) semantic option discovery (heuristic / VLM), (2) option-scoped variable-horizon diffusion, and (3) learned β for event-driven replanning in one IL hierarchy.

Experimental Setup

Push-T: end-effector pushes a T-block into goal (green). Obs: $2 \times 96 \times 96$ RGB; actions: 2D displacements. 200 demos labeled with **3 heuristic options**; arrows show motion.



Evaluation

- **Kernel ($k=5-9$):** longer chunks than DP $\Rightarrow$ sweep receptive field.
- β : bottleneck vs. obs; completion vs. +escape; expert vs. DDIM teacher; vs. duration stop.
- **SODA vs. DP:** fix k , T_a , termination; fair head-to-head (Results $\rightarrow$).

Architecture

Execution Loop

1. π_{high} classifies the current observation into an option ω (semantic sub-task).
2. π_{low} generates a variable-horizon action chunk conditioned on (s, ω) and executes T_a steps.
3. β evaluates the current state at each step. If it fires: replan within ω (perturbation) or query π_{high} for a new option (sub-goal complete).

High-Level Policy π_{high}

ResNet-18 encoder + MLP classifier trained on *all* demonstration frames to predict the expert's current option ω . Training on all states (not just segment starts) ensures π_{high} generalises to mid-skill and recovery states encountered at inference.

Low-Level Policy π_{low}

Columbia Diffusion Policy (1D U-Net + ResNet-18) extended with:

- **Option conditioning:** Embed($\omega, d=32$) concatenated to the global FiLM conditioning vector.
- **Duration channel:** $D+1$ action output; extra channel predicts normalised segment length for *temporal stretching* at inference.
- **Termination head β :** MLP on bottleneck or encoded observation. Trained with BCE.

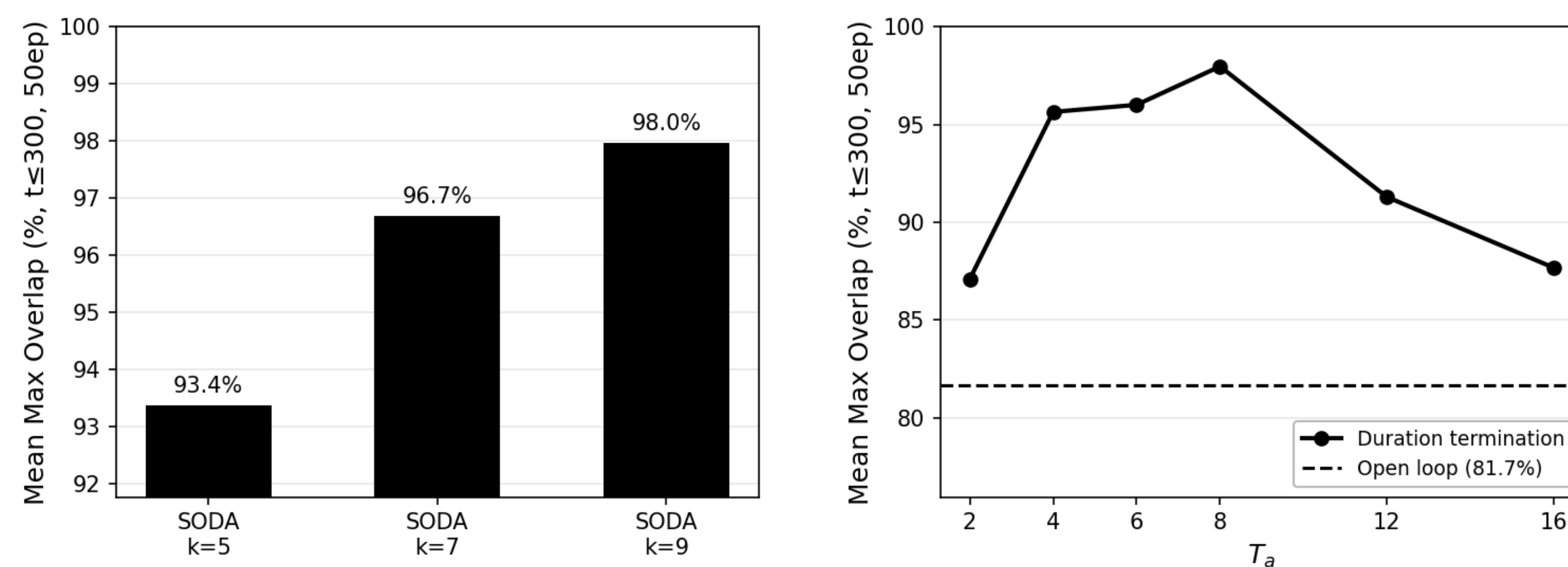
Termination

Call π_{high} every step and change option when $\beta_{\text{high}} > \text{threshold}$. Call π_{low} every step and replan chunk when $\beta_{\text{low}} > \text{threshold}$.

Results

What Drives SODA Performance?

Left: U-Net kernel size — larger receptive field improves option-conditioned diffusion (avg segment ~ 17 steps). *Right:* receding-horizon window ($T_a=8$ is optimal).



Kernel: best epoch per k , 50 episodes, $t \leq 300$ — monotonic gain as receptive field grows with longer option segments. T_a : sweep at $k=9$, duration termination; dashed = open-loop baseline.

Results

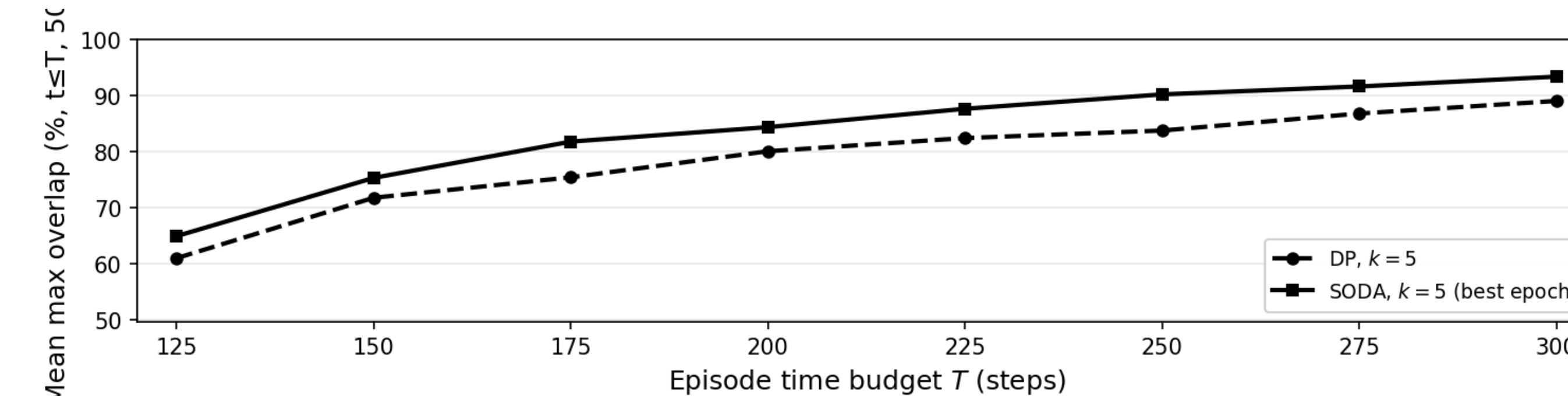
Termination Study

Learned termination only degrades performance vs. **duration termination** ($k=9$, $T_a=8$, 50 episodes).

	Duration termination	Bottleneck expert completion	Bottleneck DDIM completion	Bottleneck DDIM completion + escape	Observation completion	Observation completion + escape
Frozen β (stop-grad)	98.0	89.0	84.1	78.3	86.0	74.9
Joint β (stage 3)	—	87.3	89.6	73.5	84.2	71.7

SODA vs. Diffusion Policy

Fair $k=5$ comparison ($T_a=8$, duration termination, $T \leq 300$, 50 ep): **SODA 93.4%** vs. **DP 89.0%** (+4.3 pp). *Below:* mean max overlap vs. time budget T — SODA leads DP at every timestep.



Insights, Limitations & Future Work

- **Insight:** SODA needs a wider U-Net receptive field than flat DP because option segments average ~ 17 steps — performance rises monotonically $k=5 \rightarrow 9$.
- **Insight:** receding horizon is not “always larger T_a ”: $T_a=8$ maximises overlap; open-loop execution collapses without periodic replanning.
- **Insight:** learned β (bottleneck or observation; completion or +escape; expert or DDIM teacher) consistently trails **duration termination** on Push-T.
- **Insight:** hierarchy helps even at matched kernel: fair $k=5$, $T_a=8$, duration stop — SODA **93.4%** vs. DP **89.0%** (+4.3 pp), and SODA leads at every time budget T .
- **Limitation:** single task (Push-T) and **heuristic** option labels; VLM (Square) and LOVE (unsupervised) pipelines built but not yet evaluated here.
- **Limitation:** termination ablation missing some cells (bottleneck expert +escape); best SODA uses $k=9$ while the fair DP baseline is $k=5$.
- **Limitation:** cannot yet isolate hierarchy from kernel scaling without a DP $k=9$ control and without testing BID under the same protocol.
- **Future:** Square 6-DOF manipulation with VLM-supervised options (E2); LOVE unsupervised discovery on Push-T and Square (E3/E4).
- **Future:** β as a standalone ResNet-18 decoupled from π_{low} ; BID under the same eval protocol as SODA/DP.
- **Planned (final report):** finish β grid (bottleneck expert +escape); DP $k=9$ control to separate hierarchy from kernel scaling.
- **Planned (final report):** observation-noise eval ($\eta=0$ vs. 1) for DP, SODA, and BID; perturbation-heavy rollout reporting.